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Simulation of temperature control and irrigation time in the production of tulips using Fuzzy logic

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Abstract

We are now living in the Intelligent Industry or Industry 4.0 era; therefore, it is very important its application in situations that allow us to simulate and model any necessary process. In this case, we are concerned with tulip production in a greenhouse. Tulip flowering in Holland is one of the most beautiful spring phenomena and everybody in the world is eager to reproduce it. The purpose of this research project is to simulate how to control the temperature and irrigation time in a greenhouse for tulip production in the city of Ilo, Moquegua (Peru). To this end, fuzzy logic and LabView software version 16 from National Instrument were used to simulate temperature control and irrigation time in the production, activating and deactivating the chiller when the sensor informs of any deviation above or below the set temperature (15°C) keeping it constant for the process, as well as for the irrigation variable telling the system when and for how long plants should be irrigated at high, medium or low irrigation frequencies. The expected results were obtained in the simulation to control greenhouse temperature and irrigation time, since the chiller operates activating or deactivating the system to stay at the set temperature and the irrigation time is controlled depending on humidity.

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1. Introduction

Industry 4.0 or 4th Industrial Revolution are some of the terms used to describe the implementation of "smart" devices that can communicate autonomously along the value chain. In this context, technologies like Cyber-physical systems (CPS) self-organize, monitor processes and create a virtual copy of the real world, Internet of Things (IoT) connects machines, objects and people in real time and Cloud Computing offers solutions in addition to enabling the exchange and management of information, allowing production and business processes to be combined to create value for organizations, [1], [2], [3] & [4].

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Acronyms

AI	Artificial Intelligence
LabView	Laboratory Virtual Instrument Engineering Workbench
ICT	Information and communication technology
NI	National Instruments
CPPS	Cyber-physical production systems
GB	Giga bytes
VI	Virtual instruments
Fuzzy	A system of theories used in mathematics, computing, and philosophy to deal with statements that are neither true nor false.
Matlab	Matrix Laboratory

Symbols

G	Graphic
°C	Temperature in Celsius degrees
K	Temperature in Kelvin degrees

The use of the right methods and methodologies for enhancing agriculture supply chains performance is still a challenge, thus the concept of Industry 4.0 has evolved and adapted to agriculture 4.0 in order to analyse the behaviors and performance in this specific domain. Industry 4.0 also manifests in the farming sector, where the wave of Agriculture 4.0 provides multiple opportunities for farmers, consumers and the associated stakeholders, [6], [7], [4] & [5].

Agriculture 4.0 upgrades traditional production methods and world agriculture strategies to an optimized value chain using a range of emerging technologies that enhance disruptive solutions at all stages of the agricultural production chain, [5]. The introduction of interconnected new technologies and the concept of circular economy (CE) present numerous challenges, it has proved its value in the industrial sector to achieve a sustainability target, [8]. Thanks to recent advances in remote sensing and Artificial Intelligence (AI), we can now quantify field scale phenotypic information accurately and integrate the big data into predictive and prescriptive management tools, [9]. The versatility of Artificial Intelligence (AI) in process systems is not restricted to modelling and control only, but also as estimators to estimate the unmeasured parameters as an alternative to the conventional observers and hardware sensors; Today, the manufacturing industry is aiming to improve competitiveness by integrating capabilities such as sensing, communication, big data analytics, and cloud computing technologies in order to secure a new growth engine, [10]. In work carried out by [11], an advanced on–off control method based on fuzzy logic is proposed for maintaining thermal comfort in residential buildings. Due to the time-lag of the control systems and the late building thermal response, an anticipative control mechanism is required to reduce energy loss and thermal discomfort. A genetic algorithm for instrumentation control and optimization was developed using the LabVIEW graphical programming environment. The usefulness of this methodology for the optimization of a closed loop control instrument is demonstrated with minimal complexity and the programming is presented in detail to facilitate its adaptation to other LabVIEW applications, [12].

One of the world’s most beautiful and sought-after flowers are tulips thanks to their simplicity, symmetry and variety of hues, which make them the perfect house decoration and gift for friends and loved ones.

This study proposes an advanced on-off control system based on fuzzy logic to maintain greenhouse temperature and humidity at a setpoint. Due to the temporary discrepancy in control systems and delayed thermal response, an anticipated control mechanism is required to reduce energy loss and thermal variation; this is where fuzzy logic controllers allow for better performance. In order to model the simulation, we used the software LabView, which is minimally complex in terms of programming and is presented in detail to make its adaptation to other applications easier.

The fuzzy logic controller has gained a wider scope of action in processes that are difficult for mathematical modeling, since the design and tuning of these controllers are solely based on the experience of the expert in charge of the process. By knowing and applying the fuzzy logic concepts, it is possible to achieve the goal of designing a

fuzzy logic controller to control greenhouse temperature and humidity for tulip production. There is no doubt that fuzzy control is a very simple tool to apply and design for all type of processes. The fuzzy logic enables us to describe the behavior of a system using simple if-then syntax to reflect the experience of the human operator with the system. However, a fuzzy system will only respond to those entries for which rules have been programmed.

Greenhouse climate control is a complex procedure due to the number of involved variables and their interdependency. This study represents a contribution to integrate key climate parameters inside the greenhouse, which promotes a comfortable microclimate for plant growth, while saving energy and water resources. An intelligent fuzzy logic control system is introduced and improved using specific measures for temperature and humidity correlation. This study includes the following sections: literature review, material, methodology, results and conclusions for modeling and simulation of temperature and humidity control in a greenhouse for tulip production in the city of Ilo, Moquegua (Peru). The city of Ilo is located on the shores of the Pacific Ocean in southern Peru, having a minimum annual temperature of 23°C and a maximum temperature of 36°C, i.e., a very hot climate. The objective, apply fuzzy logic and the LabView software to program the moment in which the chilling system should operate and which conditions of temperature deviation from the Setpoint should trigger the system in the greenhouse. Apply fuzzy logic and the LabView software to program the moment in which the irrigation system should operate depending on greenhouse humidity.

2. Bibliographic review

2.1 The tulip

Tulipa spp is a geophyte plant, i.e., it grows underground structures (bulbs) that helps them thrive in harsh climates [13]. Each tulip variety has specific characteristics in terms of color, size and position of the flower, length and features of the stem, and leaf position, being the flower size and stem length the most valued commercial characteristics. Chart 1. At the time of cutting the flowers, the size of the buds is measured from their basis to the top and grouped under three categories: small (below 4.6 cm), medium-size (4.6 to 5 cm) and large (above 5 cm). Stem length is measured from the tip of the bud to the basis of the flower bud and grouped as short (below 37 cm), normal-size (37 to 43 cm) and long (above 43 cm), [14].

Table 1. Average stem length and size of the flower bud of four tulip varieties. Temuco, Carrillanca.1997.

Variety	Stem length (cm)	Size of flower bud (cm)
Menton	40.2	6.3
Maureen	39.4	5.5
Renown	36.6	6.1
Caravell	23.4	4.8

Source: [15]

According to [16], the protection provided by its tunic has a positive effect in the quantity of buds suitable for export (above 11 cm). Based on the data on tulip bud production and the monthly average maximum speeds registered every 12 minutes, the critical speed would be above 10.5 m/s.

Tulips are grown in cold regions, for instance the Bio Bio and Aysen regions in the South of Chile: 96% is cultivated outdoors and only 4% indoors. Their exports are mostly destined for the United States, Canada, and Holland [17].

2.2 Growing Tulips

According to [15], when growing tulip outdoors, the buds are planted in Autumn with decreasing temperatures. In this moment, a rapid grow of the roots and a slow growth of the apical bud occur, Figure 1. It is important to keep the temperature at 9°C for the first fifteen weeks. At the start of spring, flower scape and bud rapidly develop

and the anthesis occur. The flowerless bud produces only one leaf, whereas the commercial or flower bud grows more than two leaves and one flower.

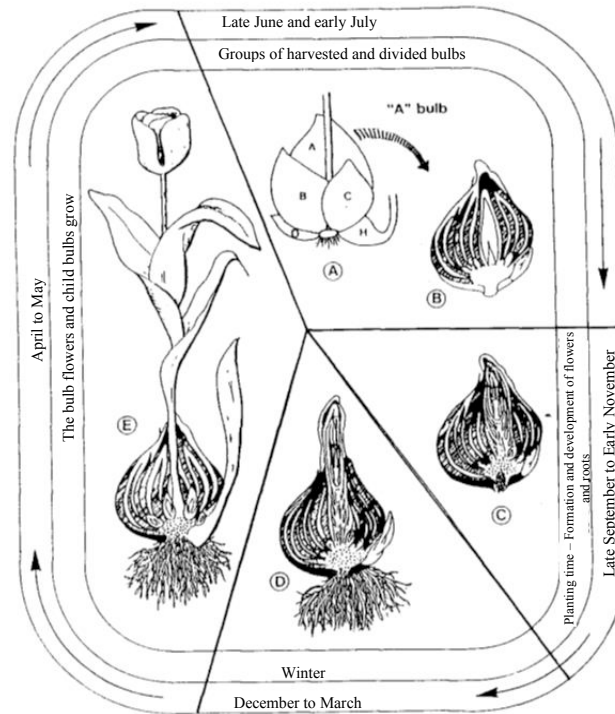


Figure 1: Annual growing and development cycle of a flower bud. (A) Harvested group; (B) Bulb “A” separated; (C) Mother bulb with root primordia and developed stems before planting; (D) Rooted bulb in a winter environment; (E) Mother bulb in anthesis (flowering), the short stem corresponds to Bulb “H”.

Source: Reported in [15].

The studies of [18] suggest splitting the nitrogen fertilization process: early in winter to promote later leaf expansion and then in Spring to cover the maximum growing period and absorption of N number of bulbs and cultivation. It is concluded that a combination of Steiner solution at 100% and fertilizing leaves with bee honey increase the quality of tulip flowers [19], [20].

The study of [16] describes weekly furrow irrigation for tulip growing. According to [15] and [21], the most widely used irrigation methods in tulip production to grow cut flower or bulbs is spray irrigation and drip tape irrigation. For [22] and [23], the average temperature for tulip growing was 15°C. As a consequence of global climate change, drought events are likely to exacerbate in terms of intensity, frequency and geographic expanse over the coming decades (IPCC, 2007) in [24]. Therefore, greenhouse growing with a controlled temperature and humidity using fuzzy logic will be important.

2.3 Agriculture and Farming 4.0

The study by [8] identifies Industry 4.0 and Circular Economy adoption barriers in the agricultural supply chain in India. Its findings will be beneficial in preparing their strategic deployment. Agriculture 4.0 is set to revolutionize agriculture productivity by using Precision Agriculture, internet of Things, Unmanned Aerial Vehicles, Internet of Underground things, and other technologies to increase agriculture produce for growing demographics while addressing various farm-related issues, [7], [6], [25]. Agriculture 4.0 upgrades traditional production methods and world agriculture strategies to an optimized value chain using a range of emerging technologies that enhance

disruptive solutions at all stages of the agricultural production chain, [5], [26]. Agriculture 4.0, as the fourth evolution in the farming technology, puts forward four essential requirements: increasing productivity, allocating resources reasonably, adapting to climate change, and avoiding food waste, [27].

The concept of Industry 4.0 has evolved and adapted to Agriculture 4.0 in order to analyze the behaviors and performance in this specific domain. Thus, the question on how Agriculture 4.0 supports a better supply chain decision-making process, or how farmers can be helped to save time when making effective decision based on objective data, remains open, according to [28]. This future research agenda provides ample scope for future interdisciplinary and transdisciplinary science on precision farming, digital agriculture, smart farming and agriculture 4.0, [29], [10].

2.4 Artificial Intelligence

According to the studies performed by [6], the concept of Data Sharing Agreement is presented as an essential path and template for AI applications of data management among various actors. This approach adopts design science principles and develops access control through a smart farm scenario while incrementally exploring the data sharing challenges in Agriculture 4.0.

Robotic Process Automation has numerous advantages in terms of automating organizational and business processes. Allied to these advantages, the complementary use of Artificial Intelligence (AI) algorithms and techniques allows to improve the accuracy and execution of processes in the extraction of information, in the recognition, classification, forecasting and optimization of processes, [4]. Emerging technologies such as computer vision and Artificial Intelligence are estimated to leverage the accessibility of big data for active training and yielding operational real time smart machines and predictable models, [30]. Forecasting accuracy is necessary for advanced process planning and for fighting uncertainty in forecasting environment. Even if there are many methods for minimizing this uncertainty by using artificial intelligence, [31]. Internet of things (IoT) and artificial intelligence (AI) are popular topics of Industry 4.0. Many publications regarding these topics have been published, but they are primarily focused on larger enterprises. However, small and medium-sized enterprises (SMEs) are considered the economic backbone of many countries, which is why it is increasingly important that these kinds of companies also have easy access to these technologies and can make them operational, [32], [33].

2.5 Fuzzy Logic

Fuzzy Logic is a method of reasoning based on approximation and assumptions that resembles the human reasoning model. Nowadays, these algorithms are widely used in fields such as robotics, production line automation, financial simulation, etc. The logic system presented by Fuzzy Logic goes after the Boolean logic when applied, and intends to assign degrees for the elements involved so that the content or non-content dichotomy does not satisfy this logic and seeks to know what is the content or non-content of a specific element, [34]. Fuzzy logic, also known as cloudy or diffuse logic, became known in 1965 when professor Lofti Zades published an article on fuzzy sets on the Information and Control Journal, thus the principles presented by this logic have existed for a long time. In the 1920s, a Polish mathematician called Jan Łukasiewicz (1878-1956) using uncertainty principles introduced the notions for the logic of vague concepts which admits a set of inaccurate values, [34], [35].

The fuzzy system is a linguistic variable whose linguistic values make up fuzzy sets. For instance, the temperature of a specific process can be a linguistic variable taking low, medium or high values. These values can be described using fuzzy sets, represented by membership functions, as shown in Figure 2, [36].

[37] designed a fuzzy logic controller to automatically adjust the relative humidity in the microwave drying process in order to improve the drying effects. By carefully building the fuzzy rules and algorithms, optimal product quality was achieved, while also accelerating the drying speed. Fuzzy based PID Controller presents better performance than PID Controller. To Model the DC Motor and for the simulation purpose LabVIEW software has been used, [38]. The monitoring systems on the climatic conditions in the agriculture industry are increasingly useful and applicable since they allow to improve the processes of growth in the plants, [39]. The efficient and sustainable irrigation mechanism results in reduced water loss and improved crop productivity, [40], [41] presents

a real-time fuzzy inference based automated irrigation controller implemented in LabVIEW, that utilizes the data logged from the sensor network and communicated using GSM/GPRS module.

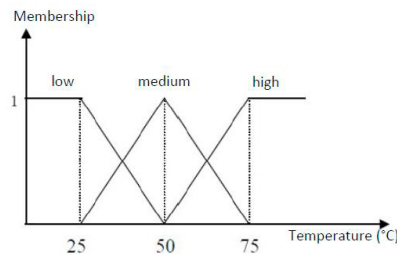


Figure 2. Membership functions for the temperature variable
Source: [36]

2.6 Simulation with LabView

[42], presented in steady the usefulness of an automated monitoring system carried out in LabView environment. The agricultural greenhouse presents a complicated procedure since the strong perturbations and the important number of its input parameters, which have a great potential and capacity to influence the climate inside it. For this reason, a Fuzzy Logic Controller (FLC) is developed in order to promote a suitable microclimate by activating the appropriate actuators installed inside the greenhouse with the appropriate rate, [43]. In the research work by [44], a simulation software on LabView is created to activate and monitor the functioning variables of a fire-tube boiler, concluding that the software is highly reliable and can be applied to similar systems. LabView is a graphical programming tool, i.e., it uses icons instead of text lines to create applications. It is highly efficient for building instrumentation, control and data acquisition systems, and has a great capability to build user interfaces for similar applications. The authors [45] developed an interface on LabView for better operation display and proved a correct interaction of the plant and the interface, reaching setpoint values with a margin of error of up to 5%. The research works by [46] deals with the modeling, simulation and supervision of a drink production and bottling pilot plant using a control system with LabView programming language, showing very good results. In the study “Herramienta de configuración y simulación de sistemas difusos tipo Mamdani para el software LabView” [Configuration Tool and Simulation of Mamdani Fuzzy Systems for the LabView Software], the Mamdani fuzzy logic was implemented since it is easy to use for the management and interpretation of diffuse systems [47].

3. Materials and methodology

3.1 Materials

The following materials were used in this research project:

- Dell Laptop, 5G of Ram and 500 GB hard drive.
- The LabView software is a graphical programming language from National Instruments.

The main scope of application of this software is measurement and automation. Programming is carried out using the data flow model, providing this language with data acquisition, monitoring and management advantages. LabView programs are virtual instruments or just VIs, and have the following components: front panel, containing the interface, and the block diagram, which includes the program graphical code. It also includes fuzzy logic and therefore, the use of MATLAB is not necessary. It is necessary to have a greenhouse implemented with basic sensors to simulate the system control with transducers to control the greenhouse temperature and irrigation time depending on humidity level. Additionally, it is necessary to have stationery.

3.2 Methodology

The problem of achieving greenhouse temperature control has traditionally been solved by using artificial intelligence techniques as the main criterion, [48]. The greenhouse climate control problem is to create a favourable environment for the crop in order to reach predetermined results for high yield, high quality and low costs, [49]. To overcome system uncertainty and parameter variations, the fuzzy logic controller is used to model the control objective based on human knowledge, understanding the system responses so there is no need for an obvious mathematical model of the system dynamics. A fuzzy controller is a system which Works on numerical data and converts it into a symbolic form through a data base (fuzzification). A logic of decision-making (rule base) is implemented, thus it is possible to provide a symbolic answer which must be converted into a numerical data (defuzzification), [49].

The design and simulation of the proposed controller used the control and simulation tool Fuzzy System Designer of the LabView Software from National Instrument, which was applied to the simulation and control of greenhouse temperature and irrigation time for tulip production on a computer through the programmed interface, following recommendations of [49], [50], and [51]. The control system design was carried out using fuzzy logic, in this case we used the model to identify the system parameters, obtaining optimal results for the controller design. The fuzzy controller technique meets the goals set in this research project, which are: high, medium or low irrigation frequencies in cold state and irrigation time, which have been verified during the real-time implementation. The controller technique employed in this project followed the recommendation by [52]. The general procedure for fuzzy controller design is as follows: Plant variables (inputs, states and outputs) are identified. The values of each variable are established. Generally, standardized values operate in the range of (0, 1) to (−1, +1). Values are then divided into a convenient number of fuzzy subsets, assigning each a linguistic label.

Assign a membership function for each fuzzy subset. Assign the relationships between the nput or state fuzzy subsets with the output fuzzy subsets, thus forming the rule base. Choose appropriate scaling factors for the input and output variables in order to standardize the variables to the (0,1) or the (−1, +1) intervals, as recommended by [53]. Establish the fuzzification and defuzzification methods to be applied. Development of the simulation and control system for greenhouse temperature and irrigation time. The research project conducted by [53] and [43] employ for the design and implementation of fuzzy controllers the development stages built based on the experimental model of the plant and the stabilization points of parameters in the system processes. The procedure followed in the stages of experimentation and validation of the results was obtained with the LabView platform to monitor controller response for comparison and experimentation purposes, using the number of membership functions of each controller as a reference. These references were the motivation behind this research project using fuzzy controllers. The proposed scheme measures the on-line sequential data of temperature from the greenhouse and humidity. LabView software programming was conducted using Virtual Instruments (VI) in order to generate the simulation and control system for greenhouse temperature and irrigation time for tulip production. A VI was created on LabView for this system, and the LabView virtual instruments from National Instrument were used from this point on. To create a Virtual Instrument, go to the Project Explorer window and left-click on “My Computer”, then click on “New”, and select “VI” in the drop-down list. By clicking “VI”, two windows will open, one with a white background and the other with a gray background (Figure 3); these windows correspond to the block diagram and the front panel. Below, we explain what you can do with these two panels before programming the system. To start with the designation of variables in the Fuzzy Logic system, we open the corresponding window (Figure 4) in LabView and create the variables of temperature and humidity control inside the greenhouse.

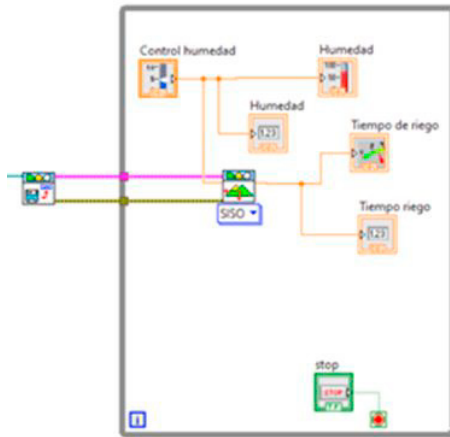


Figure 3. Block Diagram.
Source: NI LabView

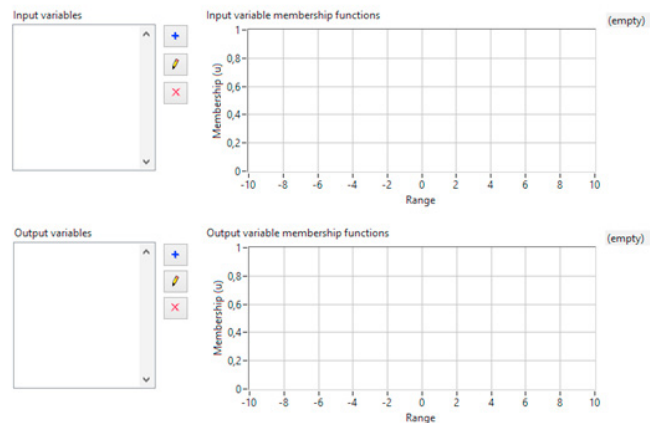


Figure 4. Fuzzy Logic system for designating
process control variables on the LabView software

The designation of temperature variables is as follows: Low Temperature, Medium Temperature and High Temperature, which are the input variables and the output variables, which refers to ventilation inside the greenhouse; low speed, medium speed and high speed. Then, the rules for these variables are setup: If the temperature in the greenhouse is low, the ventilation air will be heated until reaching the set temperature (15°C). If the temperature in the greenhouse is medium (10 to 22.5°C), the ventilation air will be at the medium temperature, and if the greenhouse temperature is high, cold ventilation air will be incorporated. Once the designation of control variables is completed, fuzzy logic elements are described next. Functional blocks are used in the design of the temperature and humidity controller in the greenhouse, first adding the FL Load Fuzzy System element (Figure 5), which allows us to upload the file from the fuzzy controller and will be used on the LabView user interface. Then, the FL FUZZY Controller is installed (Figure 6). This LabView platform tool allows to link a new fuzzy SISO system for this particular case (SISO: Single Input and Single Output system).



Figure 5. FL Load Fuzzy System

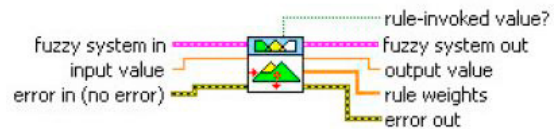


Figure 6. FL FUZZY Controller

Then, the current VI's path, strip path and build path are added to the block diagram, and all the set is combined to control the process in the greenhouse, temperature (input) and air speed (output), as well as to control humidity (input) and irrigation time (output), thus setting the rules for each of the greenhouse system variables for tulip production at a temperature and humidity set with fuzzy logic. The aim advantage of fuzzy control is the possibility of implementing human expert knowledge in the form of linguistic if – then rules. The design of a fuzzy controller begins with the choice of linguistic variables, the process state, the input and the output variables. The next step is the choice of the set of linguistic rules and the kind of fuzzy reasoning process. Once the rules are setup, after the inference, the fuzzy set and the crisp output value have to be generated; a defuzzification strategy has to be established too. The procedures used for the control and simulation of temperature and cold air speed have also been used for the control and simulation of greenhouse humidity and irrigation time, obtaining the same expected results.

4. Results and discussion

From the results obtained by working with the fuzzy inference system, it was possible to create the temperature and humidity controllers in the greenhouse for tulip production in the city of Ilo, department of Moquegua (Peru), similar to the research works by [53], [43], which present the references of the control variables, fuzzification, the system rule base and defuzzification for the controlled variables in the greenhouse, as shown in Figure 7. The control variables were designated in the Fuzzy Logic system initially for the control of greenhouse temperature: low temperature (0 to 12.5°C), medium temperature (10 to 22.5°C) and high temperature (22.5 to 40°C), as shown in Figure 8, and the output variables such as cold air speed (0 to 10°C), medium speed (10 to 19°C) and hot air speed (18 to 25°C).

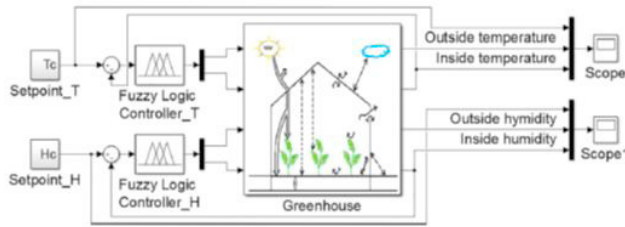


Figure 7. Model of the controlled greenhouse
Source: [43].

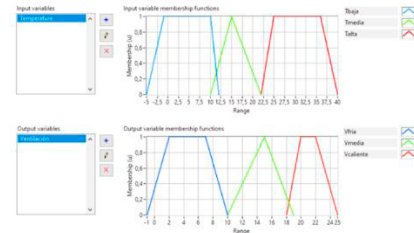


Figure 8. Greenhouse temperature and air speed control variables are entered into the system

Then, the system rules are setup (Figure 9): If the temperature in the greenhouse is low, the ventilation air will be heated until reaching the set temperature (15°C). If the temperature in the greenhouse is medium (10 to 22.5°C), the ventilation air will be set at medium temperature, and if the greenhouse temperature is high, cold ventilation air will be incorporated. Temperature control and air speed response for each variable (Figure 10) is analyzed in the system for each of the programmed variables.

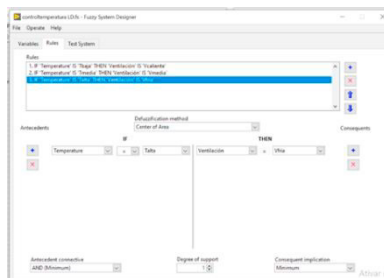


Figure 9. Rules for the control system of greenhouse temperature and air

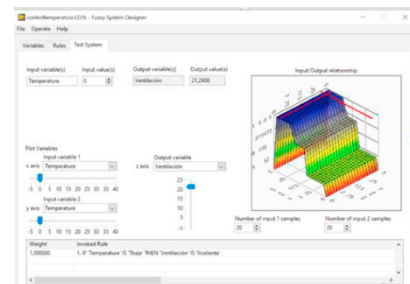


Figure 10. System analysis: if greenhouse temperature is low, then air speed will be hot (high).

Then, the block diagram programming is carried out using the links for greenhouse temperature control (Figure 11), where you can observe temperature control and air speed control. Front panel (Figure 12) shows that if greenhouse temperature is low, then the air speed is high or hot, until the temperature in the greenhouse reaches 15°C and the system is in balance.

After completion of the fuzzy logic programming for humidity (input variable) and irrigation time (output variable) Figure 13, respectively, the corresponding rules are applied, followed by the system analysis for humidity. Then, the block diagram and the respective front panel (Figure 14) are presented showing greenhouse humidity and irrigation time.

Due to the greenhouse complex dynamics of the envelopment physics, the fuzzy control represents a useful tool to be applied to this type of processes. In this paper, greenhouse temperature is controlled using basic fuzzy. Thus throughout the month including severe weather conditions the intelligent controllers are able to maintain the

programmed temperature and humidity without exact mathematical model of the greenhouse. Thereby damage of the crops and production losses are greatly reduced.

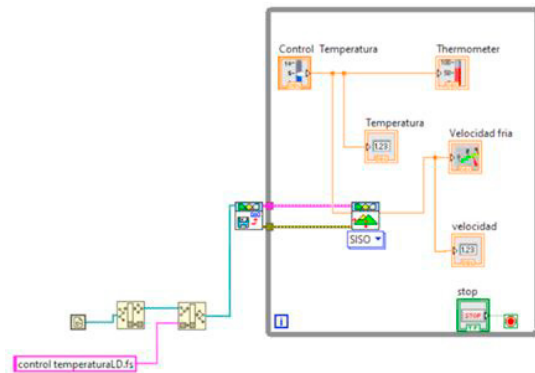


Figure 11. Block diagram with the links for greenhouse temperature control

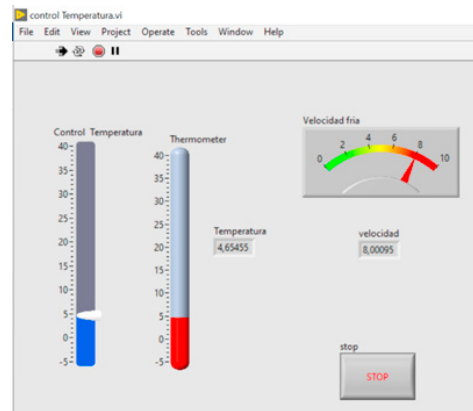


Figure 12. Front panel shows that the air speed is hot until the necessary medium temperature (15°C) is reached.

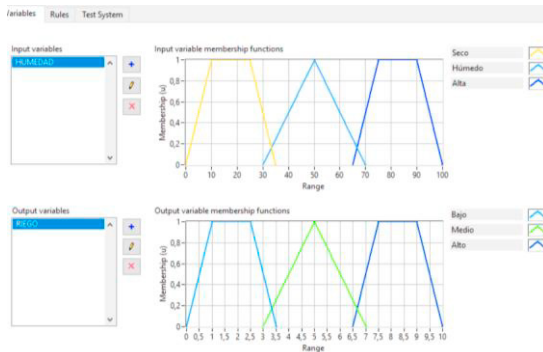


Figure 13. Greenhouse humidity and irrigation time control variables are entered into the system

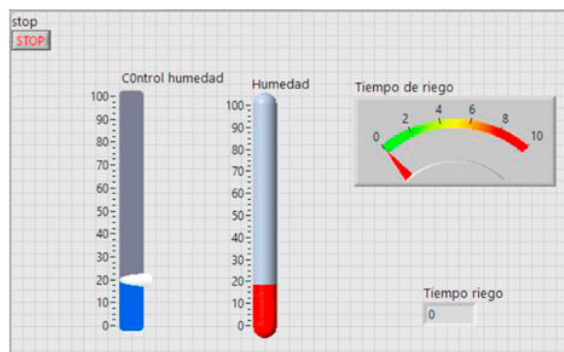


Figure 14. Front panel showing humidity and irrigation time control

The simulation results demonstrate the effectiveness of the dynamic model proposed for the research project of temperature and humidity control in the greenhouse for tulip production in the city of Ilo, Moquegua, Peru.

5. Conclusions

The most relevant feature in the fuzzy controller design is its easy implementation and good performance compared to other types of controls. Depending on the type of process to be controlled, working with this technology is very attractive. Fuzzy control proved to be very versatile because it allows for non-linear control of the system without knowing the mathematical model of the plant that describes its behavior and would be complex as demonstrated in research work [54]. Automatic control systems are widely used in various processes of different fields. The application of an automatic control system allows to improve product quality, reduce costs and production time, ensure flexibility to adapt to new products and efficient use of energy and raw materials.

Likewise, as in the conclusions of [55], Simulation results of the greenhouse dynamics illustrate the effectiveness of the proposed scheme without the exact mathematical model of the plant. The microclimate inside the greenhouse is not always suitable for plant growth. For this reason, a dynamic greenhouse model is developed

with controllers that are simulated, modeled and controlled using the fuzzy logic on LabView. The fuzzy logic controller represented a perfect tool to reach the desired microclimate inside the greenhouse, as it maintains indoor air temperature at 15 °C and a constant humidity value. In future research works, the proposed model can be applied to many other types of plants, and the developed controller can continue to be implemented to show its real-time efficiency.

A fuzzy logic control model is integrated into a hierarchical control architecture to manage crop growth. This fuzzy model is simple and very accurate, while properly regulating temperature and humidity. The simulation results show that the expected temperature can follow the daytime and night temperature patterns. A greenhouse was simulated for this research work, showing important results for industrial greenhouses. The simulation results showed that the controller can be successfully applied to control greenhouse temperature and humidity due to its efficiency and quick response time. An intelligent control system based on fuzzy logic was introduced and improved by applying specific measures for the correlation between temperature and humidity in a greenhouse for tulip production in the city of Ilo, Moquegua (Peru).

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